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COURSE NAME _____

Practical / Sessional / Project Report

Name of Student Kaushal Kumar Ray

Name of Subject Problem Solving and programming Code CSAIML110P

Branch B.Tech CSE with AI & ML Semester First

PRN 240205241023 Roll No. 29



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CERTIFICATE

This is to Certify that this report is submitted by

Mr./Ms. _____ PRN _____

Roll No. / University Roll No. _____ a student of _____ semester of the

course _____ as a part of Practical as prescribed by the University

Syllabus for the subject _____ during the Academic

year _____ and that I have instructed / guided him / her for the said work from

time to time and found him / her to be satisfactory progressive.

Signature of Instructor

Sign. of Head of Department

Registrar

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Name of Candidate : Kaushal Kumar Ray

Course Name : _____ Semester : First Subject & Code : CSAIML110P

University Roll No. : 29 P R N : 240205241023

Expt. No.	Name of Experiment / Job	Date of Expt.	Sign. of Teacher with date
01	Write a program to find the root of the given quadratic equation using Switch case	02-10-2024	
02	Write a C program to generate and print first N FIBONACCI numbers.	29-10-2024	
03	Write a C program to read two matrices and perform addition and subtraction of two matrices	12-11-2024	
04	Write a program to find whether a given number is prime or not.	19-11-2024	
05	Write a C program that reverse a given number and check whether the number is palindrome or not.	26-11-2024	
06	Write a program to find the factorial of a number using function	03-12-2024	
Total Marks of (10 x No. of Expts)			

Kaushal

Signature of Student

Signature of Faculty

[illegible]



Experiment No:1

TITLE: Program to find the Roots of a Given quadratic equation using switch case.

AIM: To write a C program to find the roots of a quadratic equation and determine if real roots exist using a switch-case statement

THEORY: The switch statement in C is a control structure used to simplify the selection between different cases based on the value of a single variable or expression. It's a more efficient and readable alternative to multiple if-else conditions when dealing with a single variable compared to multiple constant values.

Software/Hardware requirements

- Software: GCC Compiler, Text Editor
- Hardware: Any system capable of running C Compiler.

Code:

```
#include <stdio.h>
#include <conio.h>
#include <math.h>
int main ()
{
    double a, b, c, discriminant, root1, root2;
    printf("Enter coefficients a, b, c:");
    scanf("%lf %lf %lf", &a, &b, &c);
    discriminant = (b*b) - (4*a*c);
```



```
Switch (discriminant >= 0)
```

```
{
```

```
case 1:
```

```
root1 = (-b + sqrt(discriminant)) / (2*a);
```

```
root2 = (-b - sqrt(discriminant)) / (2*a);
```

```
printf("Roots are real: %.2lf and %.2lf\n", root1, root2);  
break;
```

```
case 0:
```

```
printf("No real roots possible\n");
```

```
break;
```

```
}
```

```
get ch();
```

```
return 0;
```

```
}
```

Output:

Enter coefficients a, b, c: 1

2

1

Roots are real: -1.00 and -1.00



Experiment No: 02

TITLE: Program to Generate and print the First N Fibonacci Numbers

AIM: To write a C program to generate and display the first N numbers of the Fibonacci sequence.

Software / Hardware requirement:

- Software : GCC compiler, Text editor.
- Hardware : Any system capable of running C compiler.

THEORY:

Fibonacci numbers are a sequence of numbers in mathematics where each number is the sum of the two preceding ones, starting from 0 and 1. The sequence begins as follows: 0, 1, 1, 2, 3, 5, 8, 13, ----
General Rule is : $F(n) = F(n-1) + F(n-2)$

Code:

```
#include <stdio.h>
#include <conio.h>
int main ()
{
    int n, first=0, second=1, next, i;
    printf("Enter the number of Fibonacci terms to generate:");
    scanf("%d", &n);
    printf("Fibonacci sequence : \n");
    for (i=1; i<=n; i++)
    {
```



```

if (i==1)
{ printf("%d", first); }
else if (i==2)
{ printf("%d", second); }
else
{ next = first + second;
  printf("%d", next);
  first = second;
  second = next;
}
printf("\n");
getch();
return 0;
}

```

Output:

Enter the number of Fibonacci terms to generate : 10

Fibonacci sequence :

0 1 1 2 3 5 8 13 21 34



Experiment No: 3

TITLE: program to perform Addition and subtraction of two matrices.

AIM: To write a C program to read two matrices and perform their addition and subtraction.

Software/Hardware requirements:

- Software: GCC compiler, text editor
- Hardware: Any system capable of running C compiler.

THEORY:

Here, A two-dimensional array is used. It is like a table matrices with rows and columns where each element is identified by indices with the help of rows and columns. we use basic arithmetic operators and concept of loops in this program

Code:

```
#include <stdio.h>

int main()
{
    int rows, cols, i, j;
    // Input matrix dimensions
    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &rows, &cols);
    int matrix1[rows][cols], matrix2[rows][cols], sum[rows][cols], diff[rows][cols];
    // input first matrix
    printf("Enter elements of the first matrix: \n");
```



```
for (i=0; i<rows; i++)
```

```
{
```

```
for (j=0; j<cols; j++)
```

```
{
```

```
printf("matrix1[%d][%d]: ", i, j);
```

```
scanf("%d", &matrix1[i][j]);
```

```
}
```

```
}
```

```
// Input second matrix
```

```
printf("Enter elements of the second matrix: \n");
```

```
for (i=0; i<rows; i++)
```

```
{
```

```
for (j=0; j<cols; j++)
```

```
{
```

```
printf("matrix2[%d][%d]: ", i, j);
```

```
scanf("%d", &matrix2[i][j]);
```

```
}
```

```
}
```

```
// perform addition & subtraction
```

```
for (i=0; i<rows; i++)
```

```
{
```

```
for (j=0; j<cols; j++)
```

```
{
```

```
Sum[i][j] = matrix1[i][j] + matrix2[i][j];
```

```
diff[i][j] = matrix1[i][j] - matrix2[i][j];
```

```
}
```

```
}
```




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// Display results

printf("\n Sum of the matrices: \n");

for (i=0; i<rows; i++)

{

for (j=0; j<cols; j++)

{

printf("%d", sum[i][j]);

}

printf("\n");

}

printf("\n Difference of the matrices: \n");

for (i=0; i<rows; i++)

{

for (j=0; j<cols; j++)

{

printf("%d", diff[i][j]);

}

printf("\n");

}

return 0;

}

Output:

Enter the number of rows and columns: 2

2

Enter elements of the first matrix:

matrix1[0][0]: 1

matrix1[0][1]: 2

matrix1[1][0]: 3

matrix1[1][1]: 4

Enter elements of the second matrix:

matrix2[0][0]: 5

matrix2[0][1]: 6

matrix2[1][0]: 7

matrix2[1][1]: 8

Sum of the matrices:

6 8

10 12

Difference of the matrices:

-4 -4

-4 -4



Experiment No: 4

TITLE: Program to check whether a given number is a prime number or not.

AIM: To write a C program to determine whether a given number is a prime number.

Software/Hardware requirements:

- Software: GCC compiler, Text editor
- Hardware: Any system capable of running C program

THEORY: The if-else statement allows the program to execute one block of code if the condition is true and another block of code if the condition is false. Break statement is used to stop the execution of loop or any condition inside the program.

Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int num, i, isPrime = 1;
```

```
// Input a number
```

```
printf("Enter a positive integer: ");
```

```
scanf("%d", &num);
```

```
if (num <= 1)
```

```
{
```

```
isPrime = 0; // no less than or equal to 1 are not prime
```

```
}
```


else

```
{ for (i=2; i<=num/2; i++)
```

```
{ if (num % i == 0)
```

```
{
```

```
isprime=0; // no is divisible by 'i', so not prime
```

```
break;
```

```
}
```

```
}
```

```
}
```

```
// output
```

```
if (isprime)
```

```
{ printf("%d is a prime number.\n", num); }
```

```
else
```

```
{ printf("%d is not a prime number.\n", num); }
```

```
return 0;
```

```
}
```

Output:

Enter a positive integer = 4

4 is not a prime number.



Experiment No: 5

TITLE: Program to reverse a given integer number and check if it is a palindrome

AIM: To write a C program to reverse an integer and determine whether it is a palindrome.

Software/Hardware requirements

- Software: GCC compiler, Text editor
- Hardware: Any system capable of running a C compiler.

THEORY: A palindrome is a word, number, phrase or sequence that reads the same forward and backward. We use the concept of if-else and loop. Example:
input = 123, output = 321, those numbers which are same after reversing.

Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
int num, reversed = 0, remainder, original;
```

```
// input the number
```

```
printf("Enter an integer :");
```

```
scanf("%d", &num);
```

```
original = num; // Store the original no for comparison
```

```
// Reverse the no
```

```
while (num != 0)
```

```
{
```

```

remainder = num % 10; // get the last digit
reversed = reversed * 10 + remainder; // Append it to the reversed no
num /= 10; // Remove the last digit
}
// check
if (original == reversed)
{
    printf("%d is a Palindrome number.\n", original);
}
else
{
    printf("%d is not a palindrome number.\n", original);
}
return 0;
}

```

Output:

Enter an integer: 121
121 is a Palindrome number.



Experiment No: 6

TITLE: Program to find the factorial of a number using a function with return and parameter

AIM: To write a C program that calculates the factorial of a number using a function, with the return value and passing parameter.

Software/Hardware requirements

- Software: GCC compiler, text editor
- Hardware: Any system capable of running a C compiler

THEORY: The factorial of a number is the product of all positive integers from 1 to that number. It is denoted by $n!$.

A function with return and parameter in C accepts input values (parameters), performs operations, and returns a result using the return statement.

Code:

```
#include <stdio.h>
int factorial (int n)
{
    int fact = 1;
    for (int i = 1; i <= n; i++)
    {
        fact *= i; // multiply fact with the current number i
    }
    return fact; // Return the factorial.
```

```

int main()
{
    int num;
    printf("Enter a positive integer: ");
    scanf("%d", &num);
    if (num < 0)
    {
        printf("Factorial is not defined for negative numbers. \n");
    }
    else
    {
        printf("Factorial of %d is %d \n", num, factorial(num));
    }
    return 0;
}

```

Output:

Enter a positive integer: 10
 Factorial of 10 is 3628800



Experiment No: 7

TITLE: Program to find if a character is Alphabetic, Numeric, or a special character.

AIM: To write a C program that checks whether a given character is an alphabetic character, numeric or a special character.

Software / Hardware requirements

- Software: GCC compiler, Text editor
- Hardware: Any system capable of running a C compiler.

THEORY: A special program that detects whether the input from user is a alphabet, number or a special character. We use the concept of if-else-if. The if-else-if statement are used when the user has to decide between multiple options.

Code:

```
#include <stdio.h>
#include <ctype.h> //include for character checking function
int main()
{
    char ch;
    printf("Enter a character:");
    scanf("%c", &ch);
    if (isalpha(ch))
    {
        printf("%c is an Alphabetic character.\n", ch);
    }
}
```

```
else if (isdigit(ch))
```

```
{
```

```
printf("%c is a numeric character.\n", ch);
```

```
}
```

```
else
```

```
{
```

```
printf("%c is a special character.\n", ch);
```

```
}
```

```
return 0;
```

```
}
```

Output:

① Enter a character: a
a is an Alphabetic character.

② Enter a character: 1
1 is a numeric character.

③ Enter a character: @
@ is a special character.



Experiment No. 8

TITLE: Program to compute the sum of Even numbers and the sum of odd numbers using an array

AIM: To write a C program that calculates the sum of even numbers and the sum of odd numbers from a given array:

Software / Hardware requirements

- Software : gcc compiler, text editor
- Hardware : Any system capable of running C program

THEORY: An array is a collection of elements of the same data type that are stored in contiguous memory locations. Array allow you to store multiple values under a single variable name and access them using an index.

Code:

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
int n, i;
```

```
int even_sum = 0, odd_sum = 0;
```

```
printf("Enter the number of elements: ");
```

```
scanf("%d", &n);
```

```
int arr[n];
```

```
printf("Enter %d elements: \n", n);
```

```
for (i = 0; i < n; i++)
```

```
{
```

```

scanf("%d", &arr[i]);
}
for(i=0; i<n; i++)
{
    if (arr[i] % 2 == 0)
    {
        even_sum = even_sum + arr[i];
    }
    else
    {
        odd_sum = odd_sum + arr[i];
    }
}

```

```

printf("Sum of Even numbers = %d\n", even_sum);
printf("Sum of odd numbers = %d\n", odd_sum);
return 0;
}

```

Output:

Enter the number of elements : 4

Enter 4 elements:

1

2

3

4

Sum of Even numbers : 6

Sum of odd numbers : 4



Experiment No: 9

TITLE: Program to Accept different Goods with their Number, price, and Date of purchase and Display Them using Structures

AIM: To write a C program to accept details of different goods, including their number, price, and date of purchase, using the concept of Structures and displays the entered details.

Software / Hardware requirements

- Software : GCC compiler, Text Editor
- Hardware : Any system capable of running C compiler.

THEORY:

- A structure in C is a user-defined data type that allows grouping together different data types under one name. Think of it like a container that holds multiple variables of different types, all related to specific concept.
- Each variable in the structure is known as a member of the structure.
- use struct keyword inside the main() method, followed by the name of the structure and then the name of the structure variable.

Code:

```

#include <stdio.h>
// Define the structure for goods
struct Goods {
    int number;        // unique identifier for the good
    float price;        // price of the good
    char date[11];     // Date of purchase in dd/mm/yyyy format
};

int main()
{
    int n, i;
    printf("Enter the number of goods: ");
    scanf("%d", &n);
    // Declare an array of structure
    struct Goods items[n];
    // input details
    for (i = 0; i < n; i++)
    {
        printf("\n Enter details for Good %d\n", i+1);
        printf("Number: ");
        scanf("%d", &items[i].number);
        printf("Price: ");
        scanf("%f", &items[i].price);
        printf("Date of purchase (dd/mm/yyyy): ");
        scanf("%s", items[i].date);
    }

    // Display details of all goods

```




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```
printf("\n Details of goods:\n");
printf("-----\n");
printf("Number | Price | Date of purchase\n");
printf("-----\n");

for (i=0; i<n; i++)
{
    printf("%d | %.2f | %s\n", items[i].number, items[i].price,
           items[i].date);
}

return 0;
}
```

Output:

(Input)

Enter the number of goods: 2

Enter details for good 1:

Number : 101

Price : 25.75

Date of purchase (dd/mm/yyyy): 15/12/2004

Enter details for good 2

Number : 102

Price : 50.00

Date of purchase (dd/mm/yyyy) : 16/12/2004

at put

Details of Grade:

Number	Price	Date of purchase
101	25.75	15/12/2004
102	50.00	16/12/2004



Experiment No: 10

TITLE: C program to Reverse a string without using Built-in functions (using printf and scanf)

AIM: To write a C program that reverses a string without using builtin functions, utilizing simple printf and scanf for input and output.

Software/Hardware requirements.

- Software : GCC compiler and text editors like VScode, DevC++
- Hardware : Any system capable of running a C compiler.

THEORY:

- A string in C is a sequence of characters stored in a character array and terminated by a null character.
- To output the string, you can use the printf() function together with the format specifier %s to tell C that we are now working with strings.
- C also have many useful string functions, which can be used to perform certain operations on strings.
- To use them, you must include the `<string.h>` header file in your program.

Code:

```
#include <stdio.h>
int main()
{
    char str[100], reversed[100];
    int i, length = 0;
    //input
    printf("Enter a string:");
    scanf("%s", str); // scanf does not handle spaces in strings
    // calculating the length of string
    while (str[length] != '\0')
    {
        length++;
    }
    // Reverse the string
    for (i = 0; i < length; i++)
    {
        reversed[i] = str[length - i - 1];
    }
    reversed[length] = '\0'; // Add null terminator to the reversed string
    // Display
    printf("Reversed String: %s\n", reversed);

    return 0;
}
```




Explanation:

1) Input

- The user enters a string using `scanf`. However, `scanf` does not accept spaces in input, so the input must be a single word.

2) Logic

- The length of the string is determined by iterating each character until the null terminator (`'\0'`) is reached.
- A for loop is used to reverse the string by copying characters from the end of the input string to the beginning of the reversed array.

3) Output

- The reversed string is displayed using `printf`.

Output:

Enter a string: Hello

Reversed string: olleH